

CE 515: Environmental Impact Assessment

Course Code	CE 515
Course Name	Environmental Impact Assessment
L-T-P-C	3-0-0-3
Prerequisites	NA
Intended for	UG elective (3rd and 4th Year)/PG elective (M-Tech/PhD)
Mutual Exclusion	None
Approval	50th BoA (Jul 2023)

Course Contents

- **EIA Introduction:** Introduction, definitions and concepts, rationale and historical development of EIA, Evolution of EIA; EIA at project; Regional and policy levels; EIA process in India and other countries (4 hours)
- **EIA Procedure:** Initial environmental examination, environmental impact statement, environmental appraisal, environmental impact factors and areas of consideration, Screening and scoping criteria; Rapid and Comprehensive EIA; Environmental health impact assessment; Baseline collection of data; EIA pertinent environmental factors (8 hours)
- **EIA Methodologies:** Generic steps, descriptive checklists, simple interaction matrix, stepped matrix, Networks, Overlays, uniqueness ratio, habitat evaluation system, EIA models (12 hours)
- **Impact Identification, Management and Reporting:** Impact Identification, Analysis & Prediction, Development of environment management plan; Post project monitoring ; Stakeholders consultation / Public Involvement in EIA, Mitigation, elements of mitigation, structure and element of EIA report, EIA documentation, Review process, EIA Regulations in India, Environmental Management: Preventive policy of environment, waste minimisation, conservation of water and energy, use of renewable, sources, pollution audit, pollution control strategy (10 hours)
- **Case studies:** Principles, problems and strategies and remedial actions, Applications for industrial; Water resources and irrigation projects; ports and harbours, Mining, Transportation and other projects sectors, Prediction & Assessment of Impacts on the Water Environment, Air Environment and Soil Environment (8 hours)

Text Books

1. Canter, L. W., **Environmental Impact Assessment**, 2nd Edition, McGraw-Hill, 1997.
2. Glasson J., Therivel Riki, Chadwick Andrew, **Introduction to Environmental Impact Assessment**, 4th Edition, Oxford Brookes University, 2012.

References

1. Judith, P. and Eduljee, G., **Environmental Impact Assessment for Waste Treatment and Disposal Facilities**, John Wiley & Sons, 1994.
2. Burke, G., Singh, B.R., and Theodore, L., **Handbook of Environmental Management and Technology**, 2nd Edition, John Wiley & Sons, 2000.
3. Eccleston, C.H., **Environment Impact Statements: A Comprehensive Guide to Project and Strategic Planning**, John Wiley & Sons, 2000.
4. Shrivastava A.K., Baxter Nicola, Grimm Jacob, **Environmental Impact Assessment**, APH Publishers, 2003
5. Anjaneyulu Y., Manickam Valli, **Environmental Impact Assessment Methodologies**, CRC Press 2011
6. Welford, R., **Corporate Environmental Management - Systems and Strategies**, Universities Press, 1996.
7. Whitelaw, K. and Butterworth, **ISO 14001: Environmental System Handbook**, 1997